Venkata Lakshmi Parimala Pasupuleti

AI/ML Engineer · Python / LangGraph / Azure · Large Language Model (LLM) Systems & Autonomous Agents · Responsible AI

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SUMMARY

AI/ML Engineer with 9 years of production software ownership. Built AgentProbe -- an open-source LLMOps platform: evaluation pipelines, CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, and production monitoring -- the same category as MLflow and LangSmith, built independently from production need. Built LeaseI -- an 8-agent LangGraph system with retrieval-augmented generation (RAG), tool calling, HITL guardrails, and structured data extraction, evaluated across 1,240 test cases at 93.6% precision. Responsible AI in production: AI on FHIR (HIPAA-compliant PHI system, RBAC) and Entrata PHI compliance (zero security incidents across full AWS S3 migration). Seven years at Entrata shipping enterprise Python backend systems at scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI/ML & LLMOps: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Semantic Search, Prompt Engineering, Model Evaluation, LLM-as-Judge, CI/CD Eval Gate, Regression Suites, Hallucination Detection, Guardrails, Output Validation, HITL, Agent Tracing, Production Monitoring, MLOps, LLMOps, Token Optimization, Tool Calling, Function Calling, MCP

Cloud & Enterprise AI Platforms: Azure (AI Foundry, Azure OpenAI, infrastructure), AWS (S3, EC2, RDS, Glue), Docker, Kubernetes (K8s), CI/CD (GitHub Actions), OpenAI API, Anthropic API, Groq, Ollama, Multi-provider LLM routing, Multi-modal AI (OCR/document)

Languages & Frameworks: Python, JavaScript, TypeScript, Java, SQL, PL/SQL, FastAPI, REST APIs, React.js, Flask, Angular, Node.js, Bash, HTML/CSS

Data & Infrastructure: PostgreSQL, MySQL, Oracle, RabbitMQ, Redis, Microservices, HIPAA Compliance, PHI Data Handling, Libsodium, Git, Linux, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern -- External Client Delivery

- Delivered a production LangGraph agentic system for an external client (parse → plan → retrieve → validate → conditional routing) -- owned full design-to-delivery, including build-vs-buy decisions on RAG architecture, LLM orchestration, and validation layers.
- Ran architecture reviews with client stakeholders, translating complex multi-agent orchestration concepts and model selection tradeoffs into clear technical decisions for non-specialist audiences.

Entrata (PropTech SaaS · 100K+ MAU · PHI compliance · enterprise production systems)

Sep 2018 - Aug 2024

Senior Software Engineer

- **Led AWS S3 document migration with full PHI compliance** -- Libsodium-encrypted pipelines, HIPAA-adjacent data workflows, zero security incidents across the full enterprise data migration; owned responsible AI and data governance practices throughout 7-year tenure.
- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; built Python/Bash automation pipelines eliminating 80% of manual operational workload; optimized 200+ SQL queries (30% performance gain); led microservices redesign using Docker (50% scalability improvement).
- Built and owned annual HUD income-limit pipeline (180,000 domain records, housing eligibility for 100K+ residents) -- compliance-critical data with baseline validation, reconciliation, and rollback on every release cycle; owned production reliability for 7 years.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools reducing investigation time by 40%.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated data integrity checks across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: LLMOps Evaluation & Monitoring Platform for Production AI Agents

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Built an open-source LLMOps platform -- evaluation pipelines, CI-integrated regression gate blocking quality degradation before deployment, LLM-as-judge scoring with full agent tracing, multi-trial pass@k harness, hallucination detection, and token efficiency monitoring. Same category as MLflow and LangSmith, built independently from production need.
- 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama); 15-endpoint FastAPI backend; evaluated LeaseI across 1,240 production-quality test cases.

Leasel: 8-Agent LangGraph Production System with RAG & HITL Guardrails

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent StateGraph) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Deployed an 8-agent LangGraph StateGraph in production: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation. RAG over compliance documents, OCR document ingestion, tool calling, conditional routing, HITL guardrails on regulated decisions, failure handling throughout.
- 93.6% precision, 87.9% F1, pass@1 81.7% across 1,240 test cases -- with full observability through AgentProbe eval harness.

AI on FHIR: HIPAA-Compliant Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- HIPAA-compliant AI system over structured PHI data -- RBAC access controls, responsible AI data governance, NLP-to-FHIR API translation, LLM + BERT integration; reduced manual clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017